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**Understanding Interfacial Properties between Li-Rich Layered Oxide and
Electrolyte Containing Triethyl Borate**

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ABSTRACT

Boron-containing electrolyte additives have been successfully used to improve the cyclability for Li-rich layered oxide, a hopeful cathode of high energy density lithium ion battery, but available mechanisms on their contribution are diversified. In this paper, we provide evidence to confirm the mechanism that Li-rich layered oxide is protected by a solid electrolyte interface (SEI) layer derived from boron-containing electrolyte additives. Triethyl borate (TEB), a simple boron-containing molecule, is selected as the electrolyte additive, and a representative Li-rich layered oxide, $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$, is synthesized for understanding the interfacial properties between the oxide and the electrolyte through physical and electrochemical characterizations. Cyclability tests display that the as-prepared oxide exhibits a fast capacity decrease in the standard electrolyte, 1.0 M LiPF_6 in a mixed carbonate solvent of ethyl methyl carbonate (EMC), dimethyl carbonate (DMC) and ethylene carbonate (EC) (EMC/DMC/EC = 5/2/3, in weight), with only 30% capacity retention after 150 cycles at 0.5C (1 C=250 mAh g⁻¹), which can be improved to 79% when 3% TEB is introduced. Physical characterizations demonstrate that the as-prepared oxide suffers a severe structural destruction accompanied with thick deposits from electrolyte decomposition products, but the crystal structure of the oxide is well protected by a uniform solid electrolyte interface (SEI) layer formed from the preferential oxidation of TEB.

1. INTRODUCTION

The energy density of presently available lithium ion battery cannot meet the long distance requirement of electric vehicles.¹⁻⁶ Compared with the commercialized cathodes including layered LiCoO_2 and spinel LiMn_2O_4 , Li-rich layered oxide, $x\text{Li}_2\text{MnO}_3 \cdot (1-x)\text{LiMO}_2$ ($M = \text{Co}, \text{Ni}$ and Mn), owns several advantages, including high specific capacity ($>250 \text{ mAh g}^{-1}$), higher working potential and easy preparation.^{7,8} However, the commercial application of Li-rich layered oxide is restricted by its poor cyclability due to the electrolyte decomposition and the accompanying structure destruction of the oxide.⁹⁻¹⁵

It has been known that the cyclability of Li-rich layered oxide can be improved to a great extent by doping^{16,17} and surface treatment.¹⁸⁻²⁸ However, the improvement made by these approaches is not satisfactory. For example, the capacity retention of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}]\text{O}_2$ ($M = \text{Co}, \text{Ni}$ and Mn) is improved from 46% to only 79% after 48 cycles by doping Mo^{6+} into the oxide;¹⁶ Al_2O_3 -coated $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}]\text{O}_2$ shows a high capacity retention (83.8%) after 70 cycles.¹⁹ Additionally, these approaches are complicated.

Adding additives into electrolyte for the performance improvement of lithium ion battery is cost-efficient and has been attracting much attention.^{29 - 37} Boron-containing compounds as electrolyte additives are useful to improve cyclability of high voltage cathodes including Li-rich layered oxide.^{38 - 40} However, the mechanisms that have been proposed to explain the contribution of boron-containing

electrolyte additives are diversified. For example, two diversified mechanisms have been given on the effect of a commonly used electrolyte additive, tris(trimethylsilyl) borate (TMSB). One involves the cathode film due to the preferential oxidation of TMSB, which suppresses the continuous decomposition of the electrolyte and protects cathode from structure destruction.⁴¹ The other is ascribed to the interaction between the electron-deficient boron in TMSB and PF_6^- or F^- ion. The combination of TMSB- PF_6^- improves the thermal stability of LiPF_6 salts, while the combination of TMSB- F^- reduces the formation of LiF that might increases the interfacial impedance.⁴² It is inevitable that there exists the interaction between TMSB and PF_6^- or F^- , but this interaction cannot contribute to the improved cyclability of the cathode because the amount of TMSB as an electrolyte additive is too low (less than 5%) to combine the PF_6^- ions in electrolyte completely. Usually, 1 M LiPF_6 as salt in electrolyte is needed in lithium ion battery.

It is necessary to clarify the mechanism on the contribution of boron-containing electrolyte additives for the successful application of Li-rich layered oxide as cathode of high energy density lithium ion battery. In this work, we provided evidence to confirm the mechanism that Li-rich layered oxide is protected by boron-containing electrolyte additives, which involves the formation of a protective SEI layer from the preferential oxidation of the additives. Triethyl borate (TEB), a simple boron-containing compound, which has been successfully used for the cyclability improvement of 4.5 V $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$,⁴³ and a representative Li-rich layered

oxide, $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$,⁴⁴ were selected for mechanism understanding, through theoretical calculations, electrochemical measurements and physical characterizations.

2. EXPERIMENTAL SECTION

Theoretical Calculations. Gaussian 09 program package¹² was used to calculate the highest occupied molecular orbital (HOMO) energy levels of organic molecules for a comparison on the oxidative possibility among various carbonate solvents and TEB.⁴⁵⁻⁴⁸ The molecular structures were optimized with the B3LYP method together with 6-311++G(d) level basis set.¹¹ Considering the effects of solvents, the polarized continuum model (PCM) was applied to optimize the equilibrium state structures and to investigate the bulk solvent effect with a dielectric constant of 20.5 (acetone). The frequency calculations were performed at the same level to confirm that each optimized structure is related to a stationary one. The oxidation potential (E_{ox}) was obtained as described in our previous works.^{11,12}

Sample Preparations. A representative Li-rich layered oxide, $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ (LLO), was selected and synthesized by sol-gel method.^{49,50} Typically, 1g polyvinyl pyrrolidone was dissolved in the 80 mL distilled water/ethylene glycol (1/3, in volume) under magnetic stirring. 4.901 g manganese acetate, 1.224 g nickel acetate, 1.245 g cobalt acetate and 4.875 g lithium acetate as raw materials, an oxalic acid solution (9.331 g oxalic acid dissolved in 120 mL distilled water) as a chelating agent, were then added under stirring. The solution was

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4 left under the same condition for 12 h. The resulting turbid solution was heated at
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7 90°C to obtain a thick gel that was then dried at 120°C in oven. Finally, the dried gel
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10 was kept at 450°C for 5 h and at 900°C for 12 h in muffle furnace to get the final
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12 product, denoted as LLO.
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14 Triethyl borate (TEB) (>99.5%) was purchased from J&K Scientific Ltd. 1.0 M
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16 salt lithium hexafluorophosphate (LiPF₆) in a mixed carbonate solvent of ethyl methyl
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18 carbonate (EMC), dimethyl carbonate (DMC) and ethylene carbonate (EC)
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20 (EMC/DMC/EC = 5/2/3, in weight) as the standard (STD) electrolyte, was provided
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22 by Guangzhou Tinci Material Technology Co., Ltd. TEB-containing electrolytes (1, 2,
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24 3 and 4 wt.%) were prepared by adding TEB in the STD electrolyte under the
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26 conditions as described in our previous work.⁴⁰
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32 LLO electrode with an active mass loading of 1.5 mg cm⁻³ was prepared by
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34 mixing 80 wt.% LLO, 10 wt.% acetylene black and 10 wt.% polyvinylidene fluoride
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36 (PVDF), coating the mixture on Al foil, drying, and calendaring. Li/LLO 2025-coin
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38 cells were assembled with LLO cathode, prepared electrolytes, Celgard 2025
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40 separator, and lithium foil anode in the Ar-filled glove box.
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45 **Electrochemical Measurements.** The cyclability tests of Li/LLO 2025-coin
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47 cells were performed by charging/discharging between 2.0 V and 4.8 V at 0.5 C (1 C
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49 = 250 mA g⁻¹) after three formation cycles at 0.1 C cycles under room temperature
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51 (25°C) on LAND test system (CT2001A, China). The rate capability tests were
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53 performed between 2.0 V and 4.8 V at various rate currents (0.1 C, 0.2 C, 0.5 C, 1 C,
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2C, and 4C) after three formation cycles at 0.1C. Electrochemical impedance spectroscopy was performed with frequencies ranging from 100 kHz to 0.1 Hz on a PGSTAT-30 electrochemical station (Autolab, Metrohm Co., Netherlands) at discharged state of 2.0 V.

Physical Characterizations. The cycled LLO electrodes were washed with anhydrous DMC and evacuated in vacuum oven for 3h under room temperature for physical characterizations. The crystal structure of LLO was determined by X-ray diffraction (XRD, BRUKER D8 ADVANCE, Germany) with CuK α radiation in the 2 θ range of 15-90°. Scanning electron microscopy (SEM) (JSM-6510, Japan) and Transmission electron microscopy (TEM), JEM-2100, JOEL, Japan) were employed to observe the surface morphology of LLO. X-ray photoelectron spectroscopy (XPS, ESCALAB 250) with Al K α radiation ($h\nu=1486.6$ eV) under ultra-high vacuum was employed to obtain the surface composition of cycled LLO electrodes. To determine the contents of transition metals, the cycled lithium electrodes were dissolved in acidic solution that were then analyzed by Inductively Coupled Plasma (ICP, IRIS Intrepid II XSP).⁴⁰

3. RESULTS AND DISCUSSION

Improved Cyclability of LLO by TEB. The influence of TEB content on the cyclability of LLO was evaluated by charge/discharge test in the electrolytes containing various contents of TEB. The obtained results are shown in Figure 1. It can be seen from Figure 1A that LLO cycled in the STD electrolyte shows a slow capacity

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4 decay before the 105th cycle, which mainly results from the increased interfacial
5 impedance due to the successive decomposition of the electrolyte.⁵¹ After 105 cycles,
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7 the discharge capacity drops dramatically, which should be ascribed to the structural
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9 destruction of LLO. At the 150th cycle, the discharge capacity almost disappears with
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11 a capacity retention of only 30%. Apparently, LLO exhibits poor cyclability in the
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13 STD electrolyte. The electrolyte is oxidatively decomposed when LLO is charged to
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15 over 4.2 V, yielding carbon dioxide, polymers and HF.¹² The polymers cannot form a
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17 protective film on LLO particles, while the HF causes the dissolution of transition
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19 metal ions, leading to the structure destruction of LLO.⁵²⁻⁵⁶ Fortunately, this issue can
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21 be solved by using electrolyte additives, especially when a boron-containing
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23 compound, TMSB, is used.⁴⁰
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32 Similarly to TMSB, as shown in Figure 1A, TEB can significantly improve the
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34 cyclability of LLO. When 3% TEB is applied, the capacity retention of LLO at the
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36 150th cycle is improved to 79%, which is comparable to the 73% at the 220th cycle by
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38 TMSB, although the molecular structure of TEB is simpler than TMSB.⁴⁰ Unlike most
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40 of electrolyte additives that might cause negative effect on the capacity delivery
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42 when increasing their contents in the electrolyte, increasing content of TEB does not
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44 reduce but improve slightly capacity retention of LLO. This feature is related to the
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46 ionic conductivity of the SEI layer formed by boron-containing compound. It can be
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48 found from Figure 1B that the initial charge/discharge coulombic efficiency of LLO in
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50 the TEB-containing electrolytes is lower than that in the STD electrolyte, which
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4 contributes to the formation of SEI layer. It can be noted that coulombic efficiency of
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6 LLO vibrates in the STD electrolyte, but remains stable in the TEB-containing
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8 electrolytes after 105 cycles. This difference suggests that LLO suffers structural
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10 destruction when it is cycled in the STD electrolyte, which can be prevented by using
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12 TEB as an electrolyte additive. As shown in Figure 1A, all the cells with
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14 TEB-containing electrolytes experience an increase in discharge capacity from the 1st
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16 cycle to the 2nd cycle. This phenomenon can be ascribed to the negative effect of the
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18 additive on the wettability of the carbonate-based electrolyte, which can be
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20 ameliorated by charge/discharge cycling.⁵⁷
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27 **Oxidative Preference of TEB.** The DFT calculation and dQ/dV analysis were
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29 performed to analyze the oxidative stability of TEB and STD electrolyte.⁵⁸ The
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31 calculated HOMO energies of DMC, EC, EMC, and TEB, are given in Table 1. It can
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33 be seen from Table 1 that HOMO energies are in the order of TEB>EMC>DMC>EC.
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35 TEB has a higher HOMO energy than solvent molecules, suggesting that TEB can be
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37 oxidized more easily than solvents. This preference can be more obviously identified
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39 by the oxidation potentials. The geometric structures and oxidation potentials of DMC,
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41 EC, EMC, and TEB before and after one electron oxidation are presented in Figure S1.
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43 It can be seen from Figure S1 that the oxidation potential of TEB is obviously lower
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45 than those of DMC, EC and EMC, indicating that TEB can be oxidized more easily
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47 than solvents. It can be noted from Figure S1 that the bond length of one B-O bond
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49 increases from 1.37 Å to 1.46 Å after one electron oxidation. This change suggests
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that the B-O bond will be broken after the one electron oxidation, generating $(\text{C}_2\text{H}_5\text{O})_2\text{B}$ cation and $\text{C}_2\text{H}_5\text{O}$ radical, which then are incorporated into SEI layer through subsequent oxidation or polymerization.

Figure 2 presents the initial charge and discharge profiles and corresponding differential charge capacity (dQ/dV) curves of LLO in the STD and TEB-containing electrolytes at 0.1 C. As shown in Figure 2A, LLO has semblable charge behavior in the STD and TEB-containing electrolytes:⁴⁰ the capacity at the potential below 4.5 V relates to the oxidation of transition metal ions and the lithium extraction from LiMO_2 ($M = \text{Co}, \text{Ni}$ and Mn), while that at the potential around 4.5 V is attributed to oxygen release and structure rearrangement and the lithium extraction from Li_2MnO_3 . Nevertheless, their coulombic efficiency is different: 73% in the STD electrolyte, 72%, 71%, 70%, and 69% in 1%, 2%, 3%, and 4%TEB-containing electrolytes, respectively. The Li_2O formed from Li_2MnO_3 activation during charge process causes the low coulombic efficiency in the STD electrolyte. Compared to the STD electrolyte, LLO in the TEB-containing electrolytes presents the slightly lower coulombic efficiency, which implies that TEB participates in the oxidation reaction process. As shown in Figure 2B, LLO also has similar differential charge capacity curves in the STD and TEB-containing electrolytes: the wide oxidation peaks at about 4.0 V corresponding to the oxidations of $\text{Ni}^{2+}(\text{Co}^{3+})$ to $\text{Ni}^{4+}(\text{Co}^{4+})$ and the sharp oxidation peaks at about 4.5 V corresponding to Mn^{3+} to Mn^{4+} can be identified.⁴⁰ Differently, however, the onset potential is 3.84 V in the STD electrolyte, while it becomes more positive

when TEB is applied, confirming that TEB oxidizes more easily than the STD electrolyte, which helps build the protective SEI layer on LLO surface.⁵⁹ Additionally, the sharp peak potential in the TEB-containing electrolyte shifts negatively, suggesting that lithium insertion/extraction process was accelerated by the protective SEI layer formed from the easier oxidation of TEB. This nature is similar to that of TMSB, which is able to form an ionically conductive and protective SEI layer on LLO.⁴² Considering that increasing content of TEB does not yield significantly additional improvement in capacity retention of LLO, the 3%TEB-containing electrolyte was selected for further investigations.

The protection that TEB provides for LLO can be identified by differential capacity analyses, as shown in Figure 3. After cycling, as seen from Figure 3A and B, both discharge curves of LLO show voltage decaying with the increasing cycle number, which can be ascribed to the transformation from layered to spinel phase, corresponding to the potential plateau of around 3 V. However, the voltage decaying of LLO cycled in 3%TEB-containing electrolyte is not as significant as that in the STD electrolyte. Additionally, this potential plateau disappears in the STD electrolyte but remains in 3%TEB-containing electrolyte after 150 cycles. This comparison suggests that the spinel phase will collapse completely in the STD electrolyte but can be protected by using TEB. The dQ/dV curves in Figure 3C and D display this protection more clearly. Three main reduction processes can be identified, corresponding to Li occupations in tetrahedral sites, octahedral sites with reductions

of Ni^{4+} and Co^{4+} and octahedral sites with reductions of Mn^{4+} , respectively.⁴⁰ It can be found from Figure 3C and D that all the reduction peak potentials shift and the peak intensities decrease with cycling, corresponding to the transformation of layered structure to spinel phase. The peak shifting and intensity decreasing are more serious for LLO in STD electrolyte than in 3%TEB-containing electrolyte. For example, the reduction peak Re3 in the STD system shifts from 3.22 V to 2.75 V, while LLO in 3%TEB-containing electrolyte from 3.22 V to only 3.08 V from the 5th cycle to the 105th cycle. After 150 cycles, the peak Re3 almost vanishes in the STD electrolyte but remains in 3%TEB-containing electrolyte, confirming that the serious structure destruction of LLO in the STD electrolyte and the structure maintenance that TEB provides.

Rate capability of Li-rich layered oxide is another issue that limits its commercial utilization.⁶⁰⁻⁶³ Therefore, the ionic conductivity of SEI layer is important for the capacity delivery of Li-rich layered oxide under high rate current. Figure 4 shows the rate capability of LLO electrode cycled in the STD and 3%TEB-containing electrolytes with the rate currents ranging from 0.1 C to 4 C. It can be observed that the LLO in 3%TEB-containing electrolyte shows better rate capability than in the STD electrolyte. At 4 C, for example, LLO delivers a discharge capacity of 138 $\text{mAh} \cdot \text{g}^{-1}$ in 3%TEB-containing electrolyte but only 92 $\text{mAh} \cdot \text{g}^{-1}$ in the STD electrolyte. The SEI layer is also formed from the polymers of electrolyte decomposition products, which cannot provide any protection for LLO but increase the interfacial impedance

between LLO and electrolyte, leading to the poor rate capability of LLO in the STD electrolyte. Differently, the SEI layer formed from TEB is ionically conductive, which can prevent the electrolyte from successive decomposition and reduce the interfacial impedance.⁴³

Figure 5 presents the electrochemical impedance spectra of LLO in STD and 3%TEB-containing electrolytes. The semicircle at high frequencies reflects the interfacial properties between LLO and electrolyte and the slope line at low frequencies represents the diffusion of lithium ion in LLO.⁶⁴ The interfacial impedance of electrolyte/lithium can be neglected because the redox reaction of Li/Li^+ on anode is fast, with respect to the lithium insertion/extraction in LLO. Therefore, the semicircle represents the interfacial impedance between LLO/electrolyte. After 3 cycles, the interfacial impedance of the LLO electrode cycled in 3%TEB-containing electrolyte is smaller than that in the STD electrolyte, which contributes to the improved rate capability by TEB. It can be noted from Figure 5 that the electrode in the STD electrolyte has a significant change in impedance spectra after 150 cycles, indicating that LLO suffers a serious structural deterioration. Comparatively, the impedance spectrum of the electrode after 150 cycles in 3%TEB-containing electrolyte resembles the previous one, confirming that a SEI layer formed from TEB yields a protection for LLO from structure destruction.

Evidence of the Protective SEI Layer. To confirm the existence of the SEI layer on LLO and its contribution to the protection for LLO, the morphology, crystal

structure and surface composition of the cycled LLO or lithium electrode were characterized by SEM, TEM, XRD, XPS, and ICP-MS.

The SEM and TEM images of the LLO electrodes after 150 cycles are shown in Figure 6, with a comparison of fresh one. Compared to the neat and smooth surface of the fresh LLO electrode (Figure 6 A and D), the surface of the cycled LLO electrode in the STD electrolyte (Figure 6 B and E) is apparently covered with a thick film from 0 to 200 nm, which should be ascribed to the severe oxidative decomposition of electrolyte.⁶⁵ Additionally, the well-crystallized particle in fresh electrode (Figure 6D) turns into separation (Figure 6E) after cycling in the STD electrolyte. This observation explains the serious capacity decay of LLO in the STD electrolyte. In contrast, LLO cycled in 3%TEB-containing electrolyte shows the well-preserved structure and a uniform film of about 10 nm thick, as shown in Figure 6F. This difference confirms that a protective SEI layer is formed by TEB, which restrains the decomposition of electrolyte and protects LLO from structure destruction.

XRD was used to identify the change in crystal structure of LLO after 150 cycles, as shown in Figure 7. The diffraction peaks of pristine LLO (Figure 7, fresh) are indexed well as the hexagonal α -NaFeO₂ structure with space group R-3m. While the weak peaks between 20° and 25° correspond to a Li₂MnO₃-type structure as the monoclinic unit-cell C2/m.⁶⁶⁻⁶⁸ The typical layered structure for LLO can be seen from the complete splitting of the pair reflections (006)/(102) and (018)/(110).^{69,70} Besides, the diffraction peaks for the current collector (aluminum) can also be

observed. The characteristic diffraction peaks of LLO in the pristine electrode almost disappear after cycling in the STD electrolyte, as shown in Figure 7. This significant change in XRD patterns confirms that LLO suffers serious structural destruction and the SEI layer formed from electrolyte decomposition products cannot yield any protection for LLO. Differently, the LLO cycled in the 3%TEB-containing electrolyte maintains almost the same diffractions as the pristine one, confirming that the SEI layer formed from TEB can protect LLO from structural destruction.

The surface composition of the LLO electrodes after 150 cycles was analyzed by XPS. The obtained results are shown in Figure 8. In the O 1s spectra, the peak at 529.9 eV is ascribed to the Me-O in LLO, while the peaks at 531.7 eV and 533.4 eV correspond to the C=O and C-O in polycarbonates, respectively.^{71,72} Compared with the LLO cycled in the STD electrolyte, it can be easily found that the C=O and C-O peaks are weaker, while the Me-O peak of the electrode cycled in 3%TEB-containing electrolyte is stronger. Besides, lower intensity of LiF, $\text{Li}_x\text{P}_y\text{F}_z$ and Li_xPF_y peaks in the F1s and P 2p spectra observed on the electrode cycled in the 3%TEB-containing electrolyte are in agreement with the TEM observations that thick deposits exist on LLO in the STD electrolyte but only a thin SEI layer is formed on LLO in the 3%TEB-containing electrolyte. The peaks of B-O in the B 1s spectra for the LLO cycled in 3%TEB-containing electrolyte indicate that B exists in the cycled LLO, which confirms that TEB takes part in the formation of the protective SEI layer. TEB is oxidized preferentially compared to the STD electrolyte, forming the protective SEI

layer from its oxidation decomposition products rather than those of electrolyte decomposition. Therefore, the boron in TEB is incorporated into the SEI layer and endows the SEI layer with ionic conductivity, leading to the improved rate capability of LLO. The element contents on the cycled LLO electrodes, estimated from XPS analyses, are presented in Table S1. LLO cycled in the STD electrolyte contains more P and F than in the 3%TEB-containing electrolyte, indicative of the serious decomposition of the STD electrolyte, while the B exists only on the cycled LLO in the 3%TEB-containing electrolyte, indicative of the incorporation of TEB decomposition products into the SEI layer on LLO.

The contents of transition metals deposited on lithium electrode also reflect the structural stability of LLO. ICP-MS was used to further analyze the contents of transition metals on the cycled lithium electrode. The obtained results are presented in Table 2. More transition metals, especially manganese, have deposited on lithium electrode after the cell is cycled in the STD electrolyte than in the 3%TEB-containing electrolyte, confirming that the SEI layer formed from TEB provides a protection for LLO from structure destruction.

4. CONCLUSIONS

The electrochemical and physical characterizations indicate that the improved cyclic stability of Li-rich layered oxide by using triethyl borate (TEB) as an electrolyte additive is attributed to the SEI layer. TEB is preferentially oxidized compared to the carbonate-based electrolyte, forming a thin and protective SEI layer

on the oxide, which protects the oxide from structural destruction and suppresses the decomposition of carbonate-based electrolyte. The SEI layer consists of boron-containing compounds from the oxidation decomposition of TEB, and thus exhibits ionic conductivity, leading to the improved rate capability of the oxide.

■ ASSOCIATED CONTENT

Supporting Information.

Optimized structures of EC, DMC, EMC, and TEB before and after oxidation, and the element concentrations on the surface of the LLO.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Rong, H.; Xu, M.; Xing, L.; Li, W. Enhanced Cyclability of $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ Cathode in Carbonate Based Electrolyte with Incorporation of Tris(trimethylsilyl)phosphate (TMSP). *J. Power Sources* **2014**, *261*, 148-155.
- (2) Hy, S.; Chen, Y. H.; Cheng, H. M.; Pan, C. J.; Cheng, J. H.; Rick, J.; Hwang, B. J. Stabilizing Nanosized Si Anodes with the Synergetic Usage of Atomic Layer Deposition and Electrolyte Additives for Li-ion Batteries. *Appl. Mater. Interfaces* **2015**, *7*, 13801-13807.
- (3) Chen, X. Q.; Lin, H. B.; Zheng, X. W.; Cai, X.; Xia, P.; Zhu, Y. M.; Li, X. P.; Li, W. S. Fabrication of Core-Shell Porous Nanocubic $\text{Mn}_2\text{O}_3@\text{TiO}_2$ as High-Performance Anode for Lithium Ion Battery. *J. Mater. Chem. A* **2015**, *3*, 18198-18206.
- (4) Chen, M.; Chen, D. R.; Liao, Y. H.; Zhong, X. X.; Li, W. S.; Zhang, Y. G. Layered Lithium-Rich Oxide Nanoparticles Doped with Spinel Phase: Acidic Sucrose-Assisted Synthesis and Excellent Performance as Cathode of Lithium Ion Battery. *Appl. Mater. Interfaces* **2016**, *8*, 4575-4584.
- (5) Lee, S.; Ha, J.; Choi, J.; Song, T.; Lee, J. W.; Paik, U. 3D Cross-Linked Nanoweb Architecture of Binder-Free TiO_2 Electrodes for Lithium Ion Batteries. *Appl. Mater. Interfaces* **2013**, *5*, 11525-11529.
- (6) Zhou, J.; Zhang, D.; Zhang, X.; Song, H.; Chen, X. Carbon-Nanotube-Encapsulated FeF_2 Nanorods for High-Performance Lithium-Ion Cathode Materials. *Appl. Mater. Interfaces* **2014**, *6*, 21223-21229.

- (7) Xiang, X.; Li, X.; Li, W. Preparation and Characterization of Size-Uniform $\text{Li}[\text{Li}_{0.131}\text{Ni}_{0.304}\text{Mn}_{0.565}]\text{O}_2$ Particles as Cathode Materials for High Energy Lithium Ion Battery. *J. Power Sources* **2013**, *230*, 89-95.
- (8) Xiang, X. D.; Knight, J. C.; Li, W. S.; Manthiram, A. Sensitivity and Intricacy of Cationic Substitutions on the First Charge/Discharge Cycle of Lithium-Rich Layered Oxide Cathodes. *J. Electrochem. Soc.* **2015**, *162*, A1662-A1666.
- (9) Armstrong, A. R.; Holzapfel, M.; Novak, P.; Hohnson, C. S.; Kang, S. H.; Thackeray, M. M.; Bruce, P. G. Demonstrating Oxygen Loss and Associated Structural Reorganization in the Lithium Battery Cathode $\text{Li}[\text{Ni}_{0.2}\text{Li}_{0.2}\text{Mn}_{0.6}]\text{O}_2$. *J. Am. Chem. Soc.* **2006**, *128*, 8694-8698.
- (10) Wang, C. C.; Jarvis, K.; Ferreira, P.; Manthiram, A. Effect of Synthesis Conditions on the First Charge and Reversible Capacities of Lithium-Rich Layered Oxide Cathodes. *Chem. Mater.* **2013**, *25*, 3267-3275.
- (11) Xing, L.; Wang, C.; Xu, M.; Meng, X.; Zhao, S. Theoretical Insight into Oxidative Decomposition of Propylene Carbonate in the Lithium Ion Battery. *J. Phys. Chem. B* **2009**, *113*, 5181-5187.
- (12) Xing, L.; Li, W.; Wang, C.; Gu, F. L.; Xu, M.; Tan, C. L.; Yi, J. Theoretical Investigations on Oxidative Stability of Solvents and Oxidative Decomposition Mechanism of Ethylene Carbonate for Lithium Ion Battery Use. *J. Phys. Chem. B* **2009**, *113*, 16596-16602.
- (13) Zheng, J.; Gu, M.; Xiao, J.; Zuo, P.; Wang, C.; Zhang, J. G. Corrosion/Fragmentation of Layered Composite Cathode and Related Capacity/Voltage Fading during Cycling Process. *Nano Letters* **2013**, *13*, 3824-3830.
- (14) Mohanty, D.; Kalnaus, S.; Meisner, R.; Rhodes, K.; Li, J.; Payzant, E. A; Wood,

- D.; Daniel, C. Structural Transformation of A Lithium-Rich $\text{Li}_{1.2}\text{Co}_{0.1}\text{Mn}_{0.55}\text{Ni}_{0.15}\text{O}_2$ Cathode during High Voltage Cycling Resolved by in Situ X-Ray Fiffraction. *J. Power Sources* **2013**, *229*, 239-248.
- (15) Xu, B.; Fell, C. R.; Chi, M.; Meng, Y. S. Identifying Surface Structural Changes in Layered Li-Excess Nickel Manganese Oxides in High Voltage Lithium Ion Batteries: A Joint Experimental and Theoretical Study. *Energy Environ. Sci.* **2011**, *4*, 2223-2233.
- (16) Du, J. Y.; Shan, Z. Q.; Zhu, K. L.; Liu, X. Y.; Tian, J. H.; Du, H. Y. Improved Electrochemical Performance of $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ by Doping with Molybdenum for Lithium Battery. *J. Solid State Electrochem.* **2015**, *19*, 1037-1044.
- (17) Lee, C. W.; Sun, Y. -K.; Prakash, J. A Novel Layered $\text{Li}[\text{Li}_{0.12}\text{Ni}_z\text{Mg}_{0.32-z}\text{Mn}_{0.56}]\text{O}_2$ Cathode Material for Lithium-Ion Batteries. *Electrochim. Acta* **2004**, *49*, 4425-4432.
- (18) Kim, J. S.; Johnson, C. S.; Vaughey, J. T.; Thackeray, M. M. Pre-Conditioned Layered Electrodes for Lithium Batteries. *J. Power Sources* **2006**, *153*, 258-264.
- (19) He, W.; Qian, J.; Cao, Y.; Ai, X.; Yang, H. Improved Electrochemical Performances of Nanocrystalline $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ Cathode Material for Li-ion Batteries. *RSC Adv.* **2012**, *2*, 3423-3429.
- (20) Shi, S. J.; Tu, J. P.; Tang, Y. Y.; Zhang, Y. Q.; Wang, X. L.; Gu, C. D. Preparation and Characterization of Macroporous $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}\text{O}_2$ Cathode Material for Lithium-Ion Batteries via Aerogel Template. *J. Power Sources* **2013**, *240*, 140-148.
- (21) Wu, F.; Li, N.; Su, Y.; Lu, H.; Zhang, L.; An, R.; Wang, Z.; Bao, L.; Chen, S. Can Surface Modification Be More Effective to Enhance the Electrochemical

- Performance of Lithium Rich Materials? *J. Mater. Chem.* **2012**, *22*, 1489-1497.
- (22) Wu, Y.; Murugan, A. V.; Manthiram, A. Surface Modification of High Capacity Layered $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ Cathodes by AlPO_4 . *J. Electrochem. Soc.* **2008**, *155*, A635-A641.
- (23) Wang, Q. Y.; Liu, J.; Murugan, A. V.; Manthiram, A. High Capacity Double-Layer Surface Modified $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.54}\text{Ni}_{0.13}\text{Co}_{0.13}]\text{O}_2$ Cathode with Improved Rate Capability. *J. Mater. Chem.* **2009**, *19*, 4965-4972.
- (24) Sun, Y. K.; Lee, M. J.; Yoon, C. S.; Hassoun, J.; Amine, K.; Scrosati, B. The Role of AlF_3 Coatings in Improving Electrochemical Cycling of Li-Enriched Nickel-Manganese Oxide Electrodes for Li-Ion Batteries. *Adv. Mater.* **2012**, *24*, 1192-1196.
- (25) Rosina, K. J.; Jiang, M.; Zeng, D.; Salager, E.; Best, A. S.; Grey, C. P. Structure of Aluminum Fluoride Coated $\text{Li}[\text{Li}_{1/9}\text{Ni}_{1/3}\text{Mn}_{5/9}]\text{O}_2$ Cathodes for Secondary Lithium-Ion Batteries. *J. Mater. Chem.* **2012**, *22*, 20602-20610.
- (26) Wang, D.; Wang, X. Y.; Yang, X. K.; Yu, R. Z.; Ge, L.; Shu, H. B. Polyaniline Modification and Performance Enhancement of Lithium-Rich Cathode Material Based on Layered-Spinel Hybrid Structure. *J. Power Sources* **2015**, *293*, 89-94.
- (27) Wu, C. R.; Fang, X. P.; Guo, X. W.; Mao, Y.; Ma, J.; Zhao, C. C.; Wang, Z. X.; Chen, L. Q. Surface Modification of $\text{Li}_{1.2}\text{Mn}_{0.54}\text{Co}_{0.13}\text{Ni}_{0.13}\text{O}_2$ with Conducting Polypyrrole. *J. Power Sources* **2013**, *231*, 44-49.
- (28) Zhang, X. P.; Sun, S. W.; Wu, Q.; Wan, N.; Pan, D.; Bai, Y. Improved Electrochemical and Thermal Performances of Layered $\text{Li}[\text{Li}_{0.2}\text{Ni}_{0.17}\text{Co}_{0.07}\text{Mn}_{0.56}]\text{O}_2$ via Li_2ZrO_3 Surface Modification. *J. Power Sources* **2015**, *282*, 378-384.
- (29) Rong, H.; Xu, M.; Xie, B.; Liao, X.; Huang, W.; Xing, L.; Li, W.

- Tris(trimethylsilyl) borate (TMSB) as A Cathode Surface Film Forming Additive for 5V Li/LiNi_{0.5}Mn_{1.5}O₄ Li-Ion Cells. *Electrochim. Acta* **2014**, *147*, 31-39.
- (30) Liao, X.; Huang, Q.; Mai, S.; Wang, X.; Xu, M.; Xing, L.; Liao, Y.; Li, W. Self-Discharge Suppression of 4.9 V LiNi_{0.5}Mn_{1.5}O₄ Cathode by Using Tris(trimethylsilyl)borate as An Electrolyte Additive. *J. Power Sources* **2014**, *272*, 501-507.
- (31) Contestabile, M.; Morselli, M.; Paraventi, R.; Neat, R. J. A Comparative Study on the Effect of Electrolyte/Additives on the Performance of ICP383562 Li-Ion Polymer (soft-pack) Cells. *J. Power Sources* **2003**, *119–121*, 943-947.
- (32) Abouimrane, A.; Odom, S. A.; Tavassol, H.; Schulmerich, M. V.; Wu, H.; Bhargava, R.; Gewirth, A. A.; Moore, J. S.; Amine, K. 3-Hexylthiophene as A Stabilizing Additive for High Voltage Cathodes in Lithium-Ion Batteries. *J. Electrochem. Soc.* **2012**, *160*, A268-A271.
- (33) Kang, Y. S.; Yoon, T.; Lee, S. S.; Mun, J.; Park, M. S.; Park, J. H.; Doo, S. G.; Song, I.-Y.; Oh, S. M. 1,3,5-Trihydroxybenzene as A Film-Forming Additive for High-Voltage Positive Electrode. *Electrochem. Commun.* **2013**, *27*, 26-28.
- (34) Li, Z. D.; Zhang, Y. C.; Xiang, H. F.; Ma, X. H.; Yuan, Q. F.; Wang, Q. S.; Chen, C. H. Trimethyl Phosphite as An Electrolyte Additive for High-Voltage Lithium-Ion Batteries Using Lithium-Rich Layered Oxide Cathode. *J. Power Sources* **2013**, *240*, 471-475.
- (35) Yang, L.; Markmaitree, T.; Lucht, B. L. Inorganic Additives for Passivation of High Voltage Cathode Materials. *J. Power Sources* **2011**, *196*, 2251-2254.
- (36) Tan, S.; Zhang, Z.; Li, Y.; Li, Y.; Zheng, J.; Zhou, Z.; Yang, Y. Tris(hexafluoro-iso-propyl)phosphate as An SEI-Forming Additive on Improving the Electrochemical Performance of the Li[Li_{0.2}Mn_{0.56}Ni_{0.16}Co_{0.08}]O₂ Cathode

- Material. *J. Electrochem. Soc.* **2012**, *160*, A285-A292.
- (37) Han, J. G.; Lee, S. J.; Lee, J.; Kim, J. S.; Lee, K. T.; Choi, N. S. Tunable and Robust Phosphite-Derived Surface Film to Protect Lithium-Rich Cathodes in Lithium-Ion Batteries. *Appl. Mater. Interfaces* **2015**, *7*, 8319-8329.
- (38) Zhu, Y.; Li, Y.; Bettge, M.; Abraham, D. P. Positive Electrode Passivation by LiDFOB Electrolyte Additive in High-Capacity Lithium-Ion Cells. *J. Electrochem. Soc.* **2012**, *159*, A2109-A2117.
- (39) Zheng, J.; Xiao, J.; Gu, M.; Zuo, P.; Wang, C.; Zhang, J. G. Interface Modifications by Anion Receptors for High Energy Lithium Ion Batteries. *J. Power Sources* **2014**, *250*, 313-318.
- (40) Li, J.; Xing, L.; Zhang, R.; Chen, M.; Wang, Z.; Xu, M.; Li, W. Tris(trimethylsilyl)borate as An Electrolyte Additive for Improving Interfacial Stability of High Voltage Layered Lithium-Rich Oxide Cathode/Carbonate-Based Electrolyte. *J. Power Sources* **2015**, *285*, 360-366.
- (41) Liu, Y. B.; Tan, L.; Li, L. Tris(trimethylsilyl) borate as An Electrolyte Additive to Improve the Cyclability of LiMn_2O_4 Cathode for Lithium-Ion Battery. *J. Power Sources* **2013**, *221*, 90-96.
- (42) Cai, Z. J.; Liu, Y. B.; Zhao, J. H.; Li, L.; Zhang, Y. M.; Zhang, J. Tris(trimethylsilyl)borate as Electrolyte Additive to Improve Performance of Lithium-Ion Batteries. *J. Power Sources* **2012**, *202*, 341-346.
- (43) Wang, Z. S.; Xing, L. D.; Li, J. H.; Xu, M. Q.; Li, W. S. Triethylborate as An Electrolyte Additive for High Voltage Layered Lithium Nickel Cobalt Manganese Oxide Cathode of Lithium Ion Battery. *J. Power Sources* **2016**, *307*, 587-592.
- (44) Chen, M.; Xiang, X.; Chen, D.; Liao, Y.; Huang, Q.; Li, W. Polyethylene

- Glycol-Assisted Synthesis of Hierarchically Porous Layered Lithium-Rich Oxide as Cathode of Lithium Ion Battery. *J. Power Sources* **2015**, 279, 197-204.
- (45) Bilal, S.; Bibi, S.; Ahmad, S. M.; AUHA, S. Counterpoise-Corrected Energies, NBO, HOMO–LUMO and Interaction Energies of Poly(o-aminophenol) for Ammonia Sensing by DFT Methods. *Synth. Met.* **2015**, 209, 143-149.
- (46) Xu, M. Q.; Hao, L. S.; Liu, Y. L.; Xing, L. D.; Li, W. S.; Li, B. Experimental and Theoretical Investigations of Dimethyl Acetamide (DMAc) as Electrolyte Stabilizing Additive for Lithium Ion Batteries. *J. Phys. Chem. C* **2011**, 115, 6085-6094.
- (47) Li, J. H.; Xing, L. D.; Chen, J. W.; Zhou, H. B.; Xu, M. Q.; Li, W. S. Improving High Voltage Interfacial and Structural Stability of Layered Lithium-Rich Oxide Cathode by Using A Boracic Electrolyte Additive. *J. Electrochem. Soc.* **2016**, 163, A2258-A2264.
- (48) Xing, L. D.; Vatamanu, J.; Bedrova, D.; Borodin, O.; Smith, G. Electrode/Electrolyte Interface in Sulfolane-Based Electrolytes for Li Ion Batteries: A Molecular Dynamics Simulation Study. *J. Phys. Chem. C* **2012**, 116, 23871-23881.
- (49) Xiang, X.; Knight, J.; Li, W. S.; Manthiram, A. Understanding the Effect of Co^{3+} Substitution on the Electrochemical Properties of Lithium-Rich Layered Oxide Cathodes for Lithium-Ion Batteries. *J. Phys. Chem. C* **2014**, 118, 21826-21833.
- (50) Xiang, X.; Knight, J.; Li, W. S.; Manthiram, A. Understanding the Influence of Composition and Synthesis Temperature on Oxygen Loss, Reversible Capacity, and Electrochemical Behavior of $x\text{Li}_2\text{MnO}_3-(1-x)\text{LiCoO}_2$ Cathodes in the First Cycle. *J. Phys. Chem. C* **2014**, 118, 23553-23558.
- (51) Song, B.; Liu, Z.; Lai, M. O.; Lu, L. Structural Evolution and the Capacity Fade

- Mechanism upon Long-Term Cycling in Li-Rich Cathode Material. *Phys. Chem. Chem. Phys.* **2012**, *14*, 12875-12883.
- (52) Zhang, S.S. A Review on Electrolyte Additives for Lithium-Ion Batteries. *J. Power Sources* **2006**, *162*, 1379-1394.
- (53) Ito, A.; Li, D.; Sato, Y.; Arao, M.; Watanabe, M.; Hatano, M.; Horie, H.; Ohsawa, Y. Cyclic Deterioration and Its Improvement for Li-Rich Layered Cathode Material $\text{Li}[\text{Ni}_{0.17}\text{Li}_{0.2}\text{Co}_{0.07}\text{Mn}_{0.56}]\text{O}_2$. *J. Power Sources* **2010**, *195*, 567-573.
- (54) Grey, C. P.; Yoon, W. -S.; Reed, J.; Ceder, G. Electrochemical Activity of Li in the Transition-Metal Sites of O_3 $\text{Li}[\text{Li}_{(1-2x)/3}\text{Mn}_{(2-x)/3}\text{Ni}_x]\text{O}_2$. *Electrochem. Solid State Lett.* **2004**, *7*, A290-A293.
- (55) Takamatsu, D.; Orikasa, Y.; Mori, S.; Nakatsutsumi, T.; Yamamoto, K.; Koyama, Y.; Takeoshi, M.; Tatsumi, H.; Hajime, T.; Hajime, A., et.al. Effect of An Electrolyte Additive of Vinylene Carbonate on the Electronic Structure at the Surface of A Lithium Cobalt Oxide Electrode under Battery Operating Conditions. *J. Phys. Chem. C* **2015**, *119*, 9791-9797.
- (56) Yamamoto, K.; Minato, T.; Mori, S.; Takamatsu, D.; Orikasa, Y.; Tanida, H.; Nakanishi, K.; Murayama, H.; Masese, T.; Mori, T., et.al. Improved Cyclic Performance of Lithium-Ion Batteries: An Investigation of Cathode/Electrolyte Interface via in Situ Total-Reflection Fluorescence X-ray Absorption Spectroscopy. *J. Phys. Chem. C* **2014**, *118*, 9538-9543.
- (57) Wang, Z. S.; Xing, L. D.; Li, J. H.; Li, B.; Xu, M. Q.; Liao, Y. H.; Li, W. S. Trimethyl Borate as An Electrolyte Additive for High Potential Layered Cathode with Concurrent Improvement of Rate Capability and Cyclic Stability. *Electrochim. Acta* **2015**, *184*, 40-46.
- (58) Xu, K. Electrolytes and Interphases in Li-Ion Batteries and Beyond. *Chem. Rev.*

- 2014**, *114*, 11503-11618.
- (59) Zhu, Y.; Casselman, M.; Li, Y.; Wei, A.; Abraham, D. Perfluoroalkyl-Substituted Ethylene Carbonates: Novel Electrolyte Additives for High-Voltage Lithium-Ion Batteries. *J. Power Sources* **2014**, *246*, 184-191.
- (60) Hong, Y. S.; Park, Y. J.; Ryu, K. S.; Chang, S. H.; Kim, M. G. Synthesis and Electrochemical Properties of Nanocrystalline $\text{Li}[\text{Ni}_x\text{Li}_{(1-2x)/3}\text{Mn}_{(2-x)/3}]\text{O}_2$ Prepared by A Simple Combustion Method. *J. Mater. Chem.* **2004**, *14*, 1424-1429.
- (61) Wei, G. Z.; Lu, X.; Ke, F. S.; Huang, L.; Li, J. T.; Wang, Z. X.; Zhou, Z. Y.; Sun, S. G. Crystal Habit-Tuned Nanoplate Material of $\text{Li}[\text{Li}_{1/3-2x/3}\text{Ni}_x\text{Mn}_{2/3-x/3}]\text{O}_2$ for High-Rate Performance Lithium-Ion Batteries. *Adv. Mater.* **2010**, *22*, 4364-4367.
- (62) Liu, B.; Zhang, Q.; He, S.; Sato, Y.; Zheng, J.; Li, D. Improved Electrochemical Properties of $\text{Li}_{1.2}\text{Ni}_{0.18}\text{Mn}_{0.59}\text{Co}_{0.03}\text{O}_2$ by Surface Modification with LiCoPO_4 . *Electrochim. Acta* **2011**, *56*, 6748-6751.
- (63) Kang, S. H.; Thackeray, M. M. Enhancing the Rate Capability of High Capacity $x\text{Li}_2\text{MnO}_3 \cdot (1-x)\text{LiMO}_2$ (M=Mn, Ni, Co) Electrodes by Li-Ni- PO_4 Treatment. *Electrochem. Commun.* **2009**, *11*, 748-751.
- (64) Lu, D. S.; Li, W. S.; Zuo, X. X.; Yuan, Z. Z.; Huang, Q. M. A Study on Kinetics of Li^+ Insertion/Desertion in $\text{Li}_x\text{Mn}_2\text{O}_4$ ($0 \leq x \leq 1$) by Electrochemical Impedance Spectroscopy. *J. Phys. Chem. C* **2007**, *111*, 12067-12074.
- (65) Huang, W.; Xing, L.; Wang, Y.; Xu, M.; Li, W.; Xie, F.; Xia, S. 4-(Trifluoromethyl)-benzonitrile: A Novel Electrolyte Additive for Lithium Nickel Manganese Oxide Cathode of High Voltage Lithium Ion Battery. *J. Power Sources* **2014**, *267*, 560-565.
- (66) Xiang, X.; Li, W. Significant Influence of Insufficient Lithium on

- Electrochemical Performance of Lithium-Rich Layered Oxide Cathodes for Lithium Ion Batteries. *Electrochim. Acta* **2014**, *133*, 422-427.
- (67) Liu, J.; Hou, M.; Yi, J.; Guo, S.; Wang, C.; Xia, Y. Improving the Electrochemical Performance of Layered Lithium-Rich Transition-Metal Oxides by Controlling the Structural Defects. *Energy Environ. Sci.* **2014**, *7*, 705-714.
- (68) Liu, J.; Liu, J.; Wang, R.; Xia, Y. Degradation and Structural Evolution of $x\text{Li}_2\text{MnO}_3 \cdot (1-x)\text{LiMn}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}\text{O}_2$ during Cycling. *J. Electrochem. Soc.* **2013**, *161*, A160-A167.
- (69) Zhu, Z.; Zhu, L. Synthesis of Layered Cathode Material $0.5\text{Li}_2\text{MnO}_3 \cdot 0.5\text{LiMn}_{1/3}\text{Ni}_{1/3}\text{Co}_{1/3}\text{O}_2$ by An Improved Co-Precipitation Method for Lithium-Ion Battery. *J. Power Sources* **2014**, *256*, 178-182.
- (70) Li, J.; Klöpsch, R.; Stan, M. C.; Nowak, S.; Kunze, M.; Winter, M.; Passerini, S. Synthesis and Electrochemical Performance of the High Voltage Cathode Material $\text{Li}[\text{Li}_{0.2}\text{Mn}_{0.56}\text{Ni}_{0.16}\text{Co}_{0.08}]\text{O}_2$ with Improved Rate Capability. *J. Power Sources* **2011**, *196*, 4821-4825.
- (71) Li, B.; Wang, Y. Q.; Rong, H. B.; Wang, Y. T.; Liu, J. S.; Xing, L. D.; Xu, M. Q.; Li, W. S. A Novel Electrolyte with the Ability to Form A Solid Electrolyte Interface on the Anode and Cathode of A LiMn_2O_4 /Graphite Battery. *J. Mater. Chem. A* **2013**, *1*, 12954-12961.
- (72) Huang, W. N.; Xing, L. D.; Wang, Y. T.; Xu, M. Q.; Li, W. S.; Xie, F. C.; Xia, S. G. 4-(trifluoromethyl)-benzonitrile: A Novel Electrolyte Additive for Lithium Nickel Manganese Oxide Cathode of High Voltage Lithium Ion Battery. *J. Power Sources* **2014**, *267*, 560-565.

Table 1 HOMO energies of DMC, EC, EMC, and TEB.

Organic molecules	HOMO energy (eV)
dimethyl carbonate (DMC)	-0.31127
ethylene carbonate (EC)	-0.32081
ethyl methyl carbonate (EMC)	-0.30867
triethyl borate(TEB)	-0.28527

Table 2 Contents of transition metal (in ppm) on lithium electrode after 150 cycles in the STD and 3% TEB-containing electrolytes.

	Mn	Ni	Co
STD	356.1	93.3	58.5
3% TEB	64.4	32.4	16.5

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Figure Captions

- Figure 1.** Cyclability (A) and coulombic efficiency (B) of LLO in the STD and TEB-containing electrolytes.
- Figure 2.** Initial charge and discharge profiles in the STD and TEB-containing electrolytes at 0.1 C (A) and the corresponding differential charge capacity curves (B).
- Figure 3.** Selected discharge profiles and corresponding differential capacity curves of LLO in the STD (A and C) and 3%TEB-containing (B and D) electrolytes at 0.5 C.
- Figure 4.** Rate capability of LLO in the STD and 3%TEB-containing electrolytes.
- Figure 5.** Electrochemical impedance spectra of the cycled LLO in the STD and 3%TEB-containing electrolytes.
- Figure 6.** SEM and TEM images of fresh (A and D) and the cycled LLO in the STD (B and E) and 3% TEB-containing (C and F) electrolytes.
- Figure 7.** XRD patterns of fresh and the cycled LLO in the STD and 3% TEB-containing electrolytes.
- Figure 8.** XPS profiles of LLO in the STD and 3% TEB-containing electrolytes at the 150th cycle.

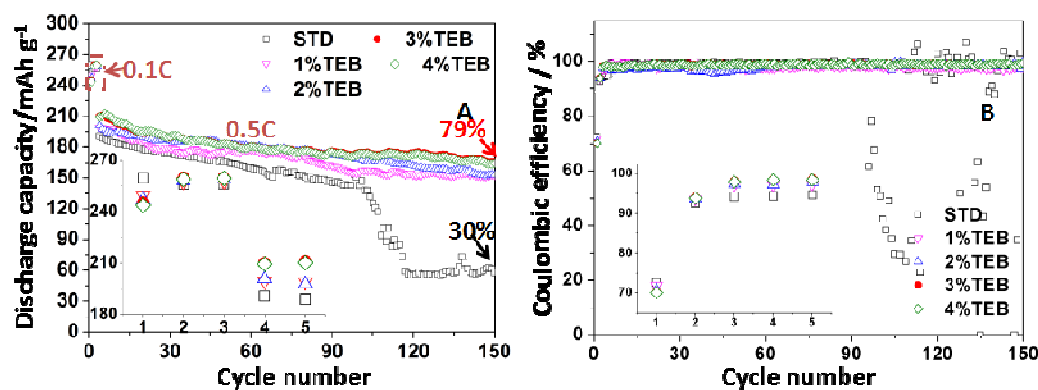


Figure 1. Cyclability (A) and coulombic efficiency (B) of LLO in the STD and TEB-containing electrolytes.

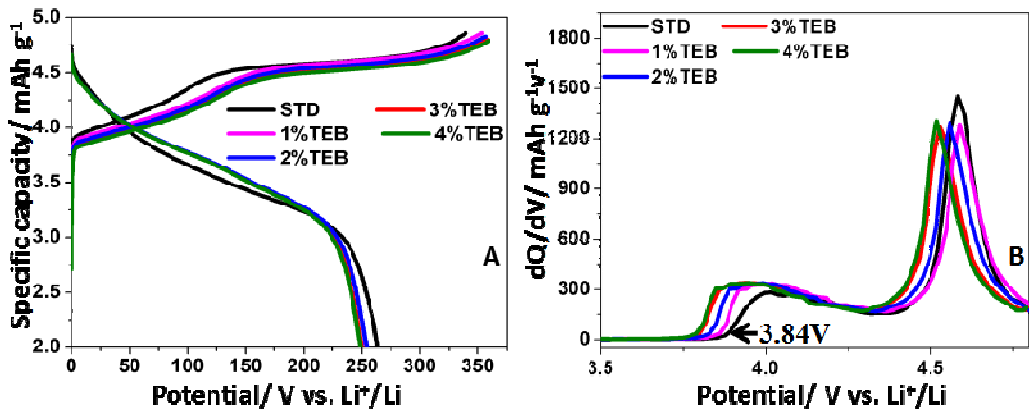


Figure 2. Initial charge and discharge profiles in the STD and TEB-containing electrolytes at 0.1 C (A) and the corresponding differential charge capacity curves (B).

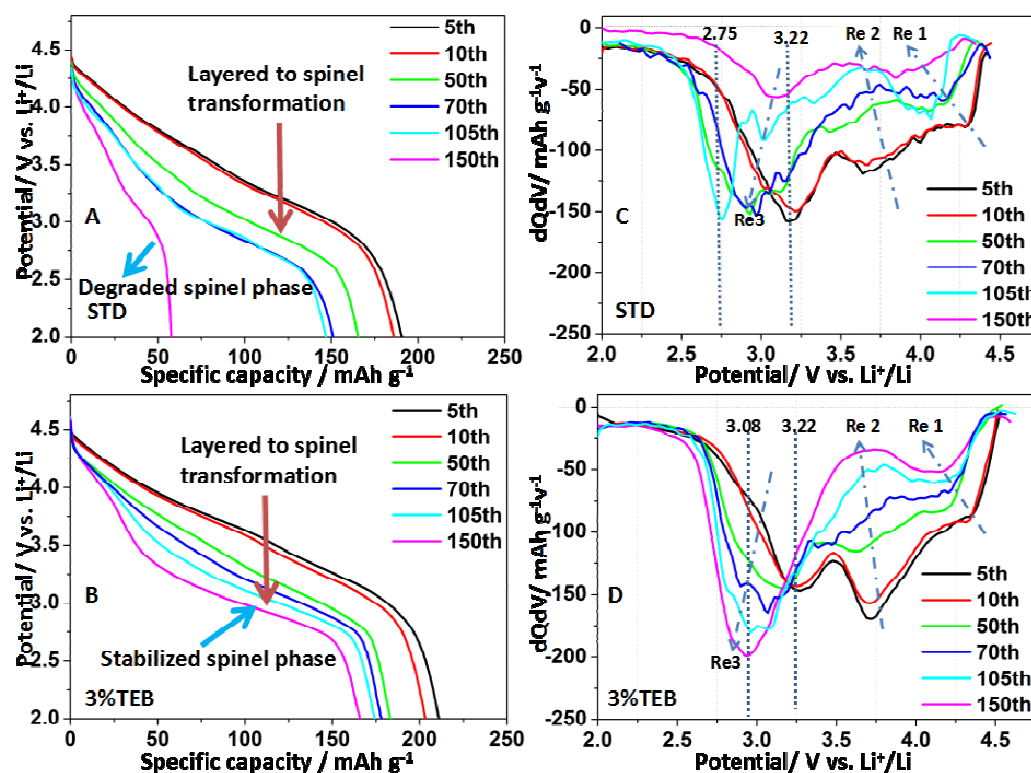


Figure 3. Selected discharge profiles and corresponding differential capacity curves of LLO in the STD (A and C) and 3%TEB-containing (B and D) electrolytes at 0.5 C.

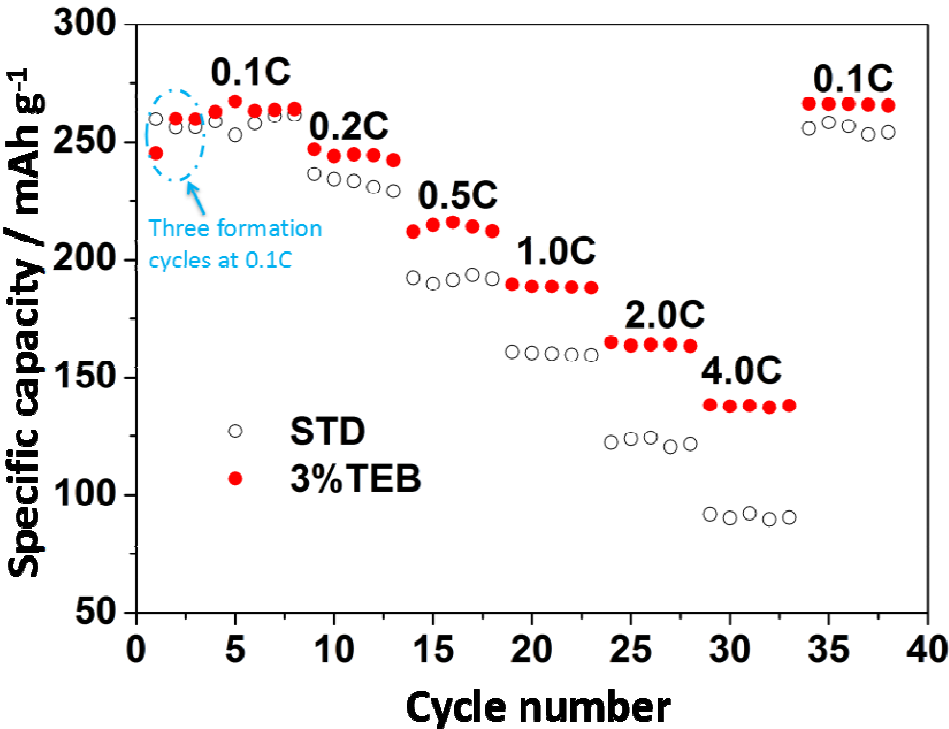


Figure 4. Rate capability of LLO in the STD and 3%TEB-containing electrolytes.

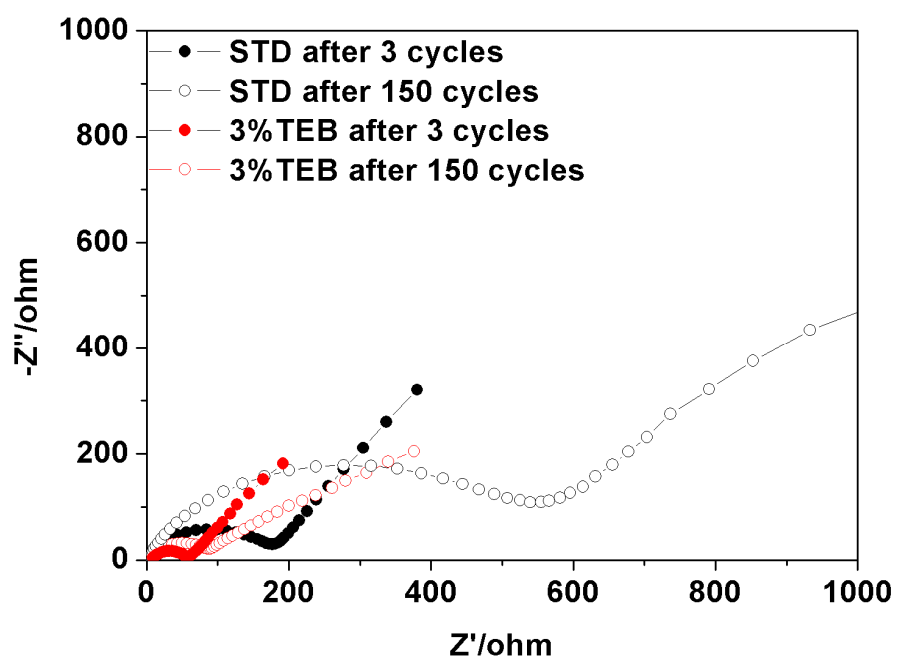


Figure 5. Electrochemical impedance spectra of the cycled LLO in the STD and 3%TEB-containing electrolytes.

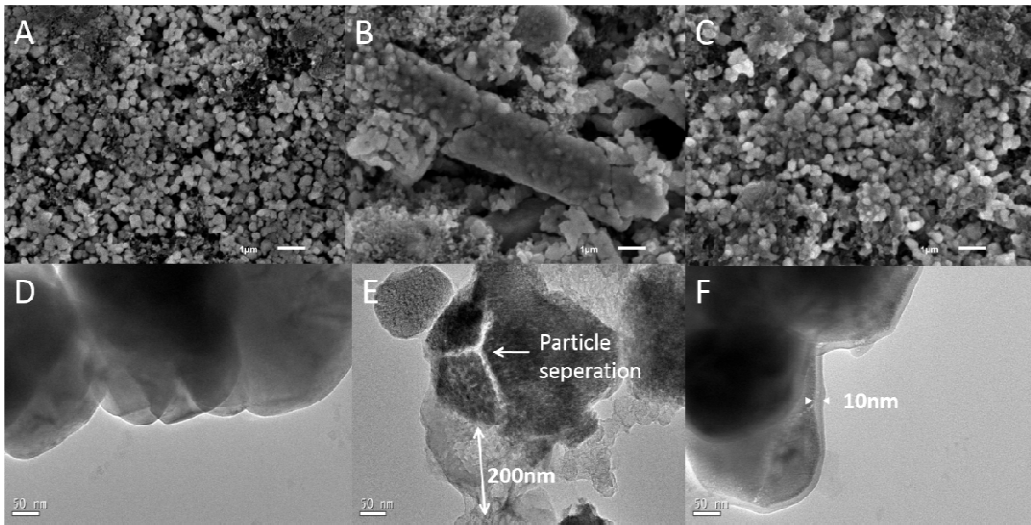


Figure 6. SEM and TEM images of fresh (A and D) and the cycled LLO in the STD (B and E) and 3% TEB-containing (C and F) electrolytes.

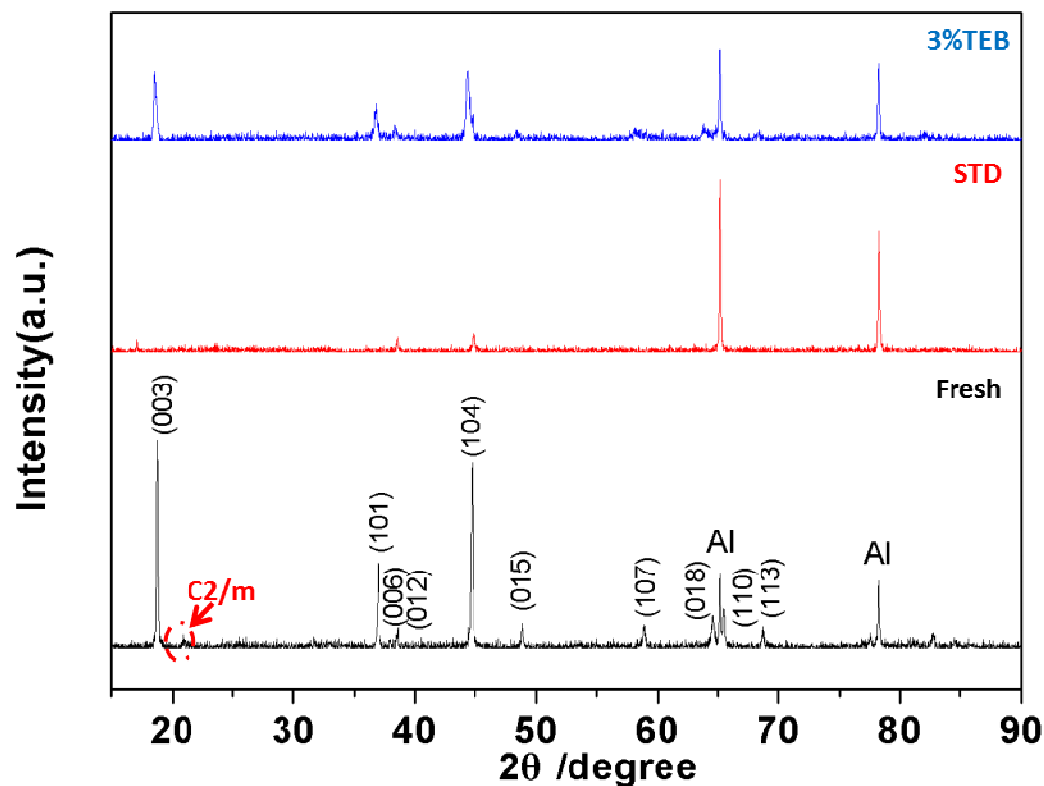


Figure 7. XRD patterns of fresh and the cycled LLO in the STD and 3% TEB-containing electrolytes.

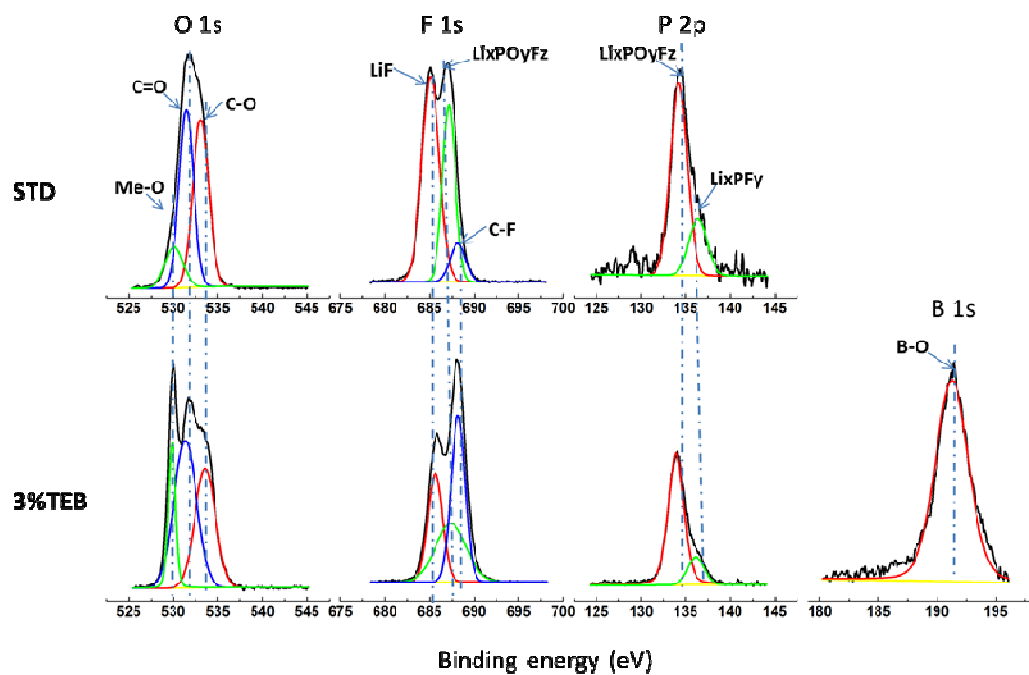


Figure 8. XPS profiles of LLO in the STD and 3% TEB-containing electrolytes at the 150th cycle.

